PRASHANTH KUMAR SIRUSALA

Data Engineer
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SUMMARY

Data Engineer with 5 years on AWS and Snowflake data pipelines for security, healthcare and advertising data, including a 17B-row Snowflake workload, Redshift tables of up to 450M rows and nightly SSIS loads from 30+ healthcare systems. Built pipelines in SQL, Python, PySpark and dbt on Airflow and Step Functions, took production on-call and led two data engineers.

TECHNICAL SKILLS

Languages & Processing: SQL, Python, PySpark (Apache Spark), Snowpark, dbt, Jinja, Airflow/MWAA, SSIS, Informatica, Fivetran, Apache Kafka, REST and GraphQL APIs, SFTP, ETL/ELT, CDC, window functions
Warehousing & Modeling: Snowflake, Amazon Redshift, PostgreSQL, SQL Server, Oracle, DB2, Apache Iceberg, data warehousing, data lakes, lakehouse, dimensional data modeling, star schemas, materialized views, data quality and reconciliation checks
AWS: Glue, Glue Data Catalog, Lambda, Step Functions, S3, Athena, Kinesis, EventBridge, SQS, SNS, DMS, AppFlow, Aurora, DynamoDB, Redshift Data API, IAM, KMS, Secrets Manager, CloudWatch
DevOps & Monitoring: Terraform, AWS CDK, Git, GitHub Actions, Bitbucket, CI/CD, Docker, Kubernetes, Snyk, Datadog
Security & Governance: HIPAA, GDPR, CCPA, PHI/PII handling, Snowflake RBAC, data masking, data retention, lineage
GenAI & BI: Claude and GPT APIs, Snowflake Cortex (LLM functions), prompt-driven SQL generation, Snowflake MCP servers, Streamlit, Tableau, Sigma, Power BI, Power Query, Salesforce Marketing Cloud
Delivery & Operations: production support, on-call, troubleshooting, root-cause analysis, performance tuning, stakeholder communication, requirements gathering, cross-functional collaboration, team leadership, technical documentation

EXPERIENCE

Data Engineer, Tech Detectives Inc., Miami, FL February 2025 – Present

- Cut average query time from 8 minutes to 1–2 minutes on a 17B-row Snowflake security-event workload by moving frequently queried attributes out of repeated raw-table scans and semi-structured parsing into incrementally refreshed tables.
- Modernized a 67-table, 1 TB legacy SQL Server and SSIS reporting stack into an S3 data lake with Apache Iceberg tables queried in Athena, fed by CDC and orchestrated by Step Functions, cutting the refresh window from 18 hours to 2 hours.
- Built two dbt projects of about 150 models each on a medallion (bronze, silver, gold) architecture, one on Snowflake and one on Redshift orchestrated by Airflow/MWAA, with incremental models, dimensional data modeling and data quality checks.
- Created reusable PySpark and SQL ingestion code for SaaS and API sources that loaded only records added since the last run.
- Implemented Kinesis and Kafka streaming ingestion for consulting clients alongside batch, API and CDC pipelines.
- Built an event-driven file pipeline on AWS in which SQS queued incoming source files, Step Functions sent each one to its own Glue job, and fact tables loaded only after every required feed arrived.
- Covered on-call for client production pipelines, 1–2 requests a day, tracing failed Glue jobs, Step Functions runs and Snowflake loads from CloudWatch and Datadog monitoring alerts to root cause and fixing most within 2–3 hours.
- Added an AI-assisted CI/CD check in which every pull request to staging ran GitHub Actions and a Lambda that called the Claude API to measure raw-to-curated ETL coverage in dev, post a report on the PR and flag anything under 90%.
- Set up Snowflake MCP servers and a chatbot that let business users query approved Snowflake data themselves, cutting ad hoc data requests to the engineering team from about 10 a week to 2.
- Led a team of two data engineers on AWS and Snowflake client work and kept building pipelines alongside them.
- Worked with analysts and business stakeholders on each engagement to gather reporting requirements and set priorities, and communicated delivery status, production incidents and trade-offs.
- Owned client AWS infrastructure as code in Terraform and AWS CDK, and locked down PII so only approved roles could read it, using least-privilege IAM, Snowflake RBAC and KMS encryption, with lineage and freshness checks on the governed datasets.
- Compared Redshift Serverless and Snowflake across 20+ source databases, recommended Snowflake because the client's BI ran on Sigma, then modeled the data and built the dbt models on Snowflake.
- Documented the AWS and Snowflake architecture, pipeline data flows, data models and production support procedures.

Data Engineer, Triangulum Insights, Remote January 2024 – February 2025

- Maintained and extended the AWS data platform (S3, Glue, Step Functions, Lambda, Redshift) for an advertising and consumer panel-data business, from raw file landing to the data warehouse tables used for reporting.
- Cut the daily refresh of the 15-table Redshift gold layer behind Tableau reporting, 100–450M rows per table, from 3 hours to 20 minutes by replacing full reloads with incremental loads and business-key deduplication.
- Performance-tuned long-running Redshift stored procedures by reading workload-management queue pressure from the system tables and rescheduling procedure runs and materialized-view refreshes away from the busiest windows.
- Automated the intake of external reporting and CRM data from outside APIs into S3 and Redshift with scheduled Python jobs.
- Designed Redshift tables and their Glue and Step Functions pipelines from the reporting team's grain, source and refresh requirements, and kept them informed during data issues and reprocessing.

- Supported more than 50 Salesforce Marketing Cloud automations and the Redshift datasets that fed them.
- Troubleshot production pipeline failures, isolating malformed files and fixing schema drift and metadata issues.
- Documented the Redshift table designs, pipeline architecture and load schedules for the reporting team.

Data Engineer, Mindtree, India January 2020 – May 2022

- Developed and maintained nightly SSIS batch integrations that moved 1–2M rows of healthcare data from 30+ source systems through SQL Server raw and staging layers into standardized reporting tables and views.
- Built a HIPAA- and GDPR-aligned retention pipeline that moved records past the 10-year policy out of healthcare source databases into an AWS archive, keeping 100M+ in-policy rows in each database.
- Gathered the retention rules from business stakeholders, agreeing which healthcare records had to be kept and which could be purged, then implemented them as the pipeline's purge and archive logic.
- Supported 15–20 nightly SSIS packages in production, reconciling source-to-target row counts before publication, tracing failed jobs to root cause and escalating source-system delays upstream.
- Applied PHI/PII masking, exclusion and access controls across the nightly healthcare pipelines.
- Documented SSIS package designs, source-to-target mappings and nightly support procedures for the healthcare workflows.

Software Engineer Intern, Mindtree, India June 2019 – December 2019

- Trained in data engineering and SQL Server Integration Services (SSIS), building ETL packages with control flows, data flow transformations, lookups and error handling to load and clean source records.
- Monitored scheduled ETL batch jobs in SQL Server Agent during on-call shifts and escalated failed or late loads to the data engineering team before reporting users saw stale or missing data.
- Handled production support with senior engineers, using SQL Server Agent job history and package error logs to find why ETL jobs failed or loaded bad data, then rerunning them once the cause was fixed.
- Resolved change requests (CRs) on SSIS packages, SQL queries and source-to-target mappings, testing each change in development and documenting it before its release to production.

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate · CompTIA Security+
- In progress: SnowPro Core, AWS Certified Data Engineer – Associate

EDUCATION

Master of Science (M.S.), Cybersecurity, Florida International University, Miami, FL

Bachelor of Technology (B.Tech), Computer Science & Engineering, Rajiv Gandhi University of Knowledge Technologies India